

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Rescheduled Supplementary Summer Examination – 2025

Course: B. Tech.

Branch : Computer Engineering /

Semester :V

Computer Science and Engineering

Subject Code & Name: BTCOC502 & Theory of Computation

Max Marks: 60

Date: 25/08/2025

Duration: 3Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is not a part of tuple finite automata? a. Input alphabet b. Transition function c. Initial State d. Output function	Understand	1
2	The number of states required by an automaton to accept string 101 are a. 2 b. 3 c. 4 d. 5	Analyze	1
3	Regular expression for string that start with ab and end with ba is a. aba^*b^*ba b. $ab(ab)^*ba$ c. $ab(a+b)^*ba$ d. All of these	Design	1
4	Regular expressions are closed under a. Union b. Concatenation c. Kleene Star d. All of these	Understand	1
5	Which grammar is in Greibach normal form (GNF) ? a. $S \rightarrow AA 0$, $A \rightarrow SS 1$ b. $A \rightarrow aB$, $B \rightarrow b$ c. $A \rightarrow aBd$, $B \rightarrow b$ d. None of these	Understand	1
6	Moore machine produces output over the change of a. transitions b. state c. input symbol d. None of these	Understand	1

7	Can Deterministic Finite automata determine palindrome number ?				Analyze	1
	a. Yes	b. NO	c.Yes for Some cases	d. yes for Z^*		
8	A grammar that produces more than one parse tree is called				Understand	1
	a. regular	b. ambiguous	c.unambiguous	d. linear		
9	The language accepted by Pushdown automata is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
10	The language accepted by Turning Machine is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
11	The Pushdown automata behaves like Turing machine if it has auxiliary memory				Understand	1
	a.0	b. 2	c.2 or more	d. None		
12	The language accepted by Finite Automata is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
Q. 2	Solve the following.					12
A)	Find the string set for following regular expressions. (i) 00^* (ii) a^*b^* (iii) $(0+1)^*$				Create & CO1	6
B)	What is FA(Finite Automaton)? Explain with examples. Elaborate on Automaton and complexity				Understand & CO1	6
Q.3	Solve the following.					12
A)	Explain in brief Chomsky hierarchy with suitable examples?				Understand & CO2	6
B)	Design a FSM to check whether a given decimal number is divisible by 5?				Understand & CO1	6
Q. 4	Solve Any Two of the following.					12
A)	Eliminate NULL production from the grammar G given below: Write the productions after elimination a. $S \rightarrow a / Xb / aYa$ b. $X \rightarrow Y / \wedge$ c. $Y \rightarrow b / X$				Apply & CO2	6

B)	Consider a left linear grammar as given below a. $S \rightarrow Sa Abc$ b. $A \rightarrow Sa Ab a$ Convert left linear grammar to right linear grammar.	Apply & CO3	6
C)	Convert the given to context free grammar to CNF forms a. $S \rightarrow ASB$ b. $A \rightarrow aAS a \epsilon$ c. $B \rightarrow SbS A bb$	Apply & CO-3	6
Q.5	Solve Any Two of the following.		12
A)	Derive a derivation tree for the string aabbabba for the given context free grammar (CFG) a. $S \rightarrow aB bA$ b. $A \rightarrow a aS bAA$ c. $B \rightarrow b bS aBB$	Analyse / CO-3	6
B)	Define a PDA & list three important properties of a PDA Machine?	Understand & CO-2	6
C)	Convert the following grammar to a PDA that accepts the same language. a. $S \rightarrow OS1 A$ b. $A \rightarrow 1A0 S \epsilon$	Apply & CO-3	6
Q. 6	Solve Any Two of the following.		12
A)	Design a TM for an equal number of a's and b's must follow a.	Create & CO-3	6
B)	Design a TM which recognizes palindromes over $= \{a, b\}$	Create & CO-3	6
C)	Explain 'Halting Problem of turning machine' with neat diagrams?	Understand & CO-4	6
	*** End ***		